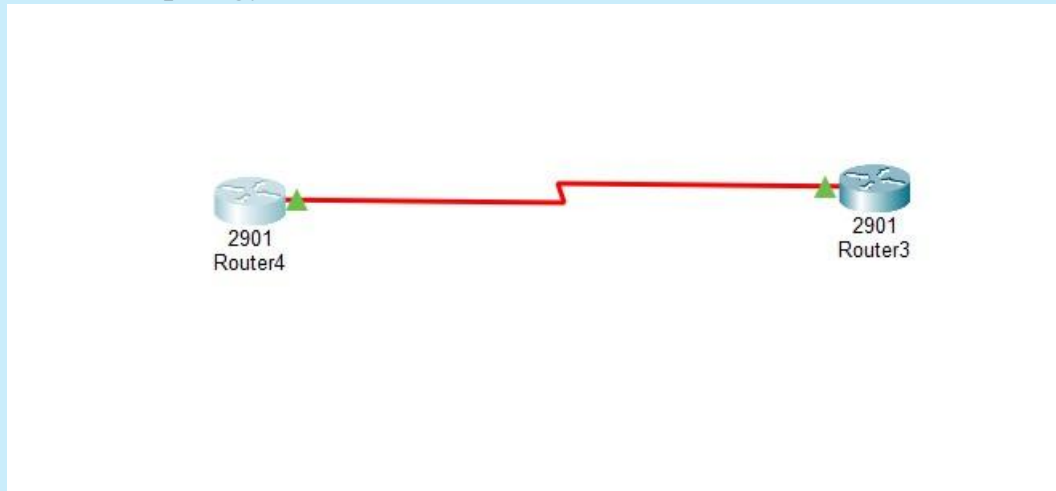


Static routing

1. Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.



Network topology



Configuration check , Connectivity test

```
Router>
Router>ena
Router#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
exit exit
      ^
% Invalid input detected at '^' marker.

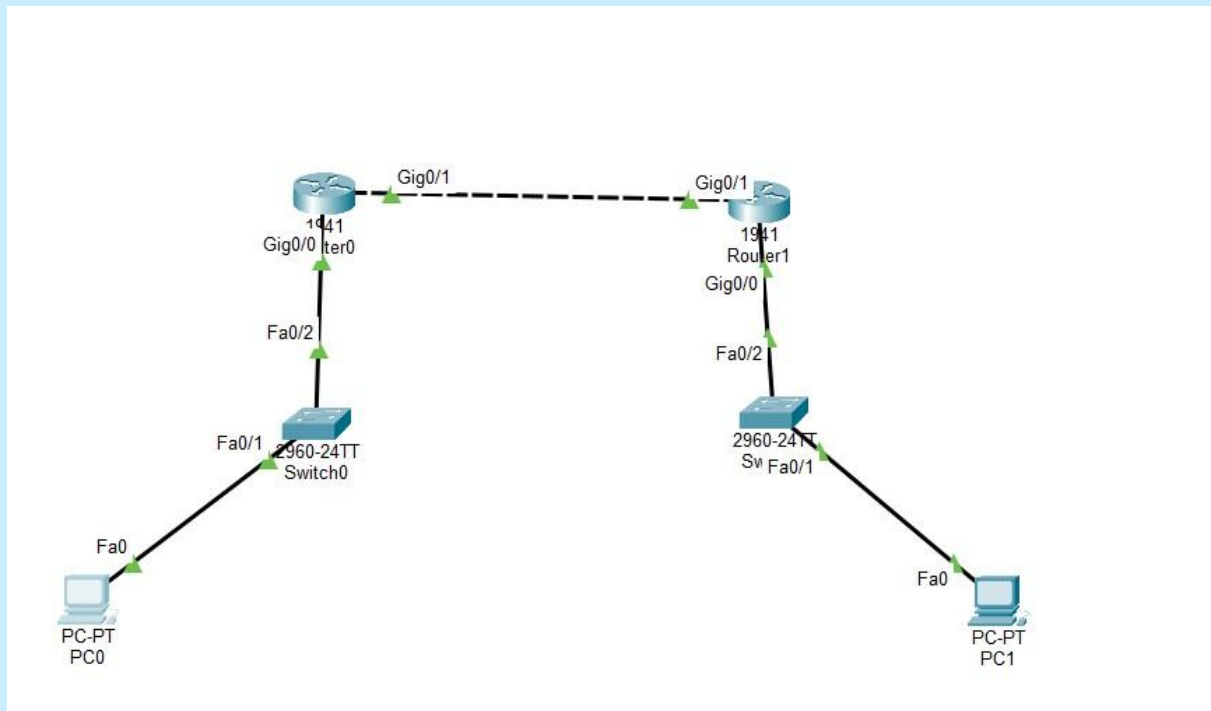
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0        unassigned      YES unset    administratively down down
GigabitEthernet0/1        unassigned      YES unset    administratively down down
Serial0/3/0               192.168.1.1     YES manual    up          up
Serial0/3/1              unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
Router#ping 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/8/31 ms
Router#
```

2. On Router1 in Cisco Packet Tracer, configure a static route so it can reach a PC connected to Router2's LAN network. Test connectivity using the ping command from Router1 to the PC.



Network topology



Configuration check , Connectivity test

```
Router>ena
Router#
Router#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip add
% Incomplete command.
Router(config-if)#ip address 192.168.10.10 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.100.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#ip route 192.168.20.0 255.255.255.0 192.168.100.2
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Test connectivity using the ping command

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time=11ms TTL=126
Reply from 192.168.20.20: bytes=32 time=14ms TTL=126
Reply from 192.168.20.20: bytes=32 time=12ms TTL=126
Reply from 192.168.20.20: bytes=32 time=10ms TTL=126

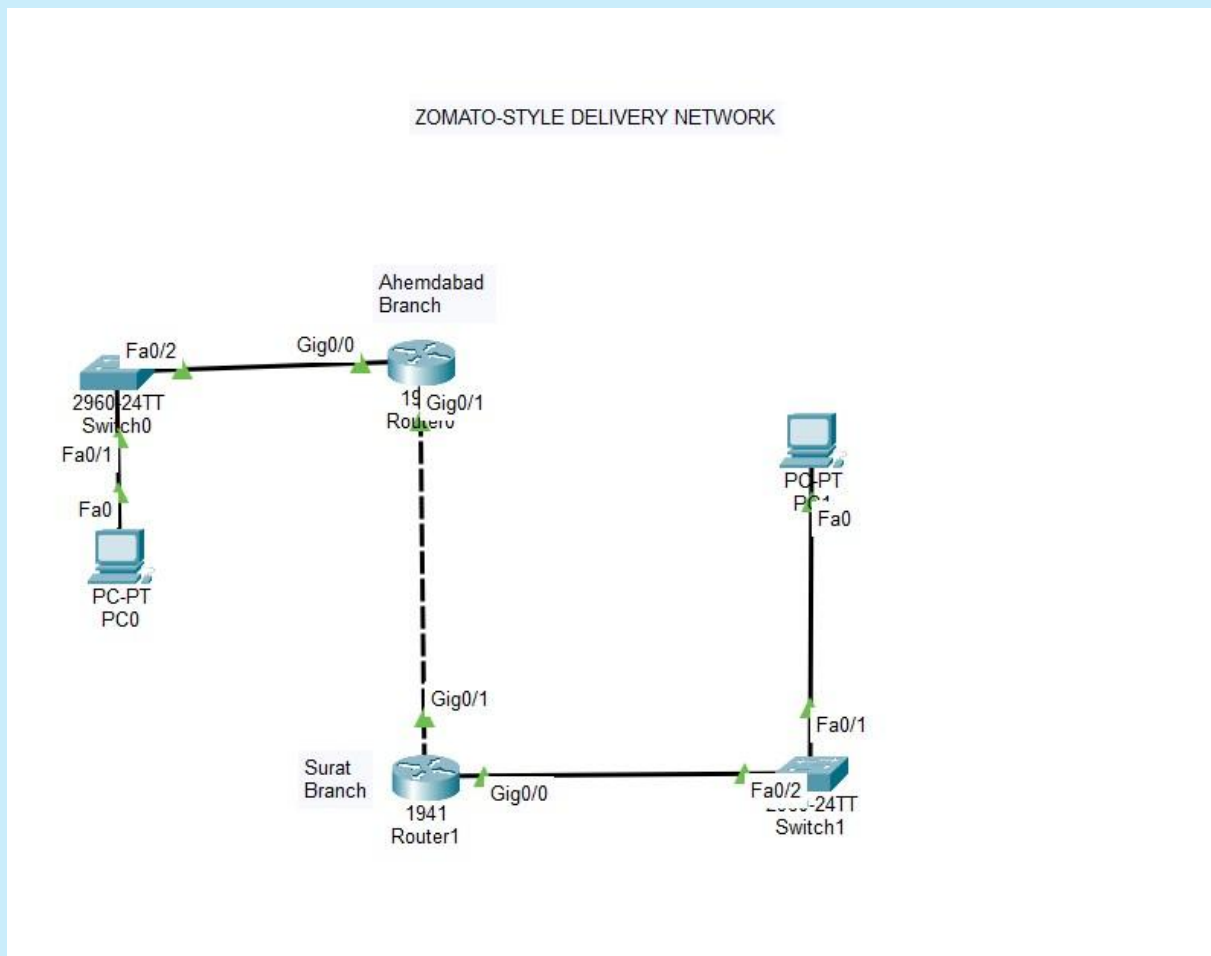
Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 14ms, Average = 11ms

C:\>
```

3. Simulate a Zomato-style delivery network: imagine Router1 represents an Ahmedabad branch and Router2 represents a Surat branch. Configure static routes so that devices in each city's LAN can communicate with each other through the routers.



Network topology



4. List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points.

Hint: Think about how static routes might impact delivery or booking updates in these apps.



Advantages and Disadvantages of Static Routing

Static routing is a method where network routes are manually configured by the administrator.

Advantages

1. **Simple and Predictable:** Static routes follow a fixed path, so the administrator knows how data will travel. For example, a small IRCTC internal network can use static routes for fixed connections between servers.
2. **Low Network Overhead:** Static routing does not exchange routing information continuously, so it uses less bandwidth and router resources.

Disadvantages

1. **Manual Updates Required:** If a network path fails, the administrator must manually change the route. For example, in Swiggy, if a network path used for delivery updates fails, order or delivery status updates may be delayed until the route is fixed.
2. **Difficult for Large Networks:** Managing many static routes becomes difficult when the network is large or frequently changing. For example, IRCTC has many systems and servers, so manually updating routes for every network change would be inefficient.